[affiliation to be completed]

Abstract

De Graaf [arXiv:2411.01199v2] has argued that dynamical arrest in two-dimensional hard disks is controlled by geometric ground states: congruent tangential tilings whose area fraction equals the isoperimetric quotient of the tile, with the floret pentagonal tiling at $\phi = \sqrt{3}\pi/7$ identified with the glass transition. We establish the three-dimensional form of that argument. For a monodisperse packing inscribed in a congruent tangential honeycomb, $\eta = 36\pi V^2/S^3 = Q_{\mathrm{iso}}$, the cube of the Wadell sphericity, so the admissibility criterion is that every Voronoi neighbour be a contact. Because the seven Voronoi face-class norms are linear functionals of the Selling parameters, tangentiality is a linear system and can be enumerated exactly: there are precisely three tangential Bravais lattices in three dimensions, and no truncated-octahedral solution exists, so BCC-derived cells are excluded structurally. Depleting FCC by a vacancy sublattice yields a discrete ladder $k \in \{4, 5, 7, 13\}$ with $\eta_k = (1 - 1/k)\pi/\sqrt{18}$, precisely those k with $(k-1) \mid 12$; the two-dimensional analogue $(k-1) \mid 6$ unifies de Graaf's floret pentagonal tiling ($k = 7$) and his honeycomb caging state ($k = 3$) as members of one family. We correct four errors in the original three-dimensional discussion. Contrary to the claim that no candidate falls in the reported glass range $\eta \approx 0.58$–0.64, three do. We then test the mechanism numerically. In a Monte Carlo pilot the proposed contact-fraction signature is found to be degenerate with the median Voronoi face gap and shows no feature at arrest, while the shell-percolation threshold $\varepsilon^*(\eta)$ is composition-independent to 4.3% across very different size distributions. We argue that $\varepsilon^*(\eta)$, not the diffusion coefficient, is the estimator on which a decisive test should be built.

Tangential Voronoi cells as geometric ground states for hard spheres: exact enumeration in three dimensions and a test of the mechanism

Felix Laga

August 16, 2026

1 Introduction

The relation between local structure and dynamical arrest in hard-particle systems remains unsettled. In two dimensions, de Graaf has recently proposed a sharp and unusual answer: that arrest is controlled not by a local structural motif in the usual sense, but by proximity to a *geometric ground state*, a congruent tiling of the plane whose associated area fraction is picked out by geometry alone. The identification rests on an exact statement. For a monodisperse packing of disks inscribed in a congruent tangential tiling, the area fraction equals the isoperimetric quotient of the tile,

$$\phi = q \equiv \frac{4\pi A}{P^2}, \tag{1}$$

so that a special value of ϕ can be read off a shape descriptor. The floret pentagonal tiling, with $\phi = \sqrt{3}\pi/7 \approx 0.777343$, is then identified with the observed glass and frictional jamming transitions, and a honeycomb arrangement at $\phi = \pi/(3\sqrt{3}) \approx 0.6046$ with the onset of caging.

The three-dimensional discussion in that work is explicitly exploratory: several candidate structures are constructed from FCC, BCC and diamond arrangements, a number of volume fractions are obtained, and the author concludes that the possible connection "should be investigated more closely."

This paper does two things. First, we close the geometry. We give the exact three-dimensional counterpart of Eq. (1), derive from it a sharp admissibility criterion, enumerate the admissible structures completely within the Bravais class, and identify a discrete family of depleted close-packed states that subsumes both of the two-dimensional structures. In the course of this we correct four errors in the original three-dimensional discussion. Second, we test the mechanism in a Monte Carlo pilot, and report a negative result for the observable that would most directly probe it, together with a positive result for a different one.

2 The three-dimensional analogue of $\phi = q$

Let a monodisperse packing of spheres of diameter σ be such that every Voronoi cell is a *tangential* polyhedron whose insphere is the particle itself. Decomposing the polyhedron into pyramids of height $\sigma/2$ erected on each face gives $V = \sigma S/6$, with V the cell volume and S its surface area. Hence

$$\eta = \frac{(\pi/6)\sigma^3}{V} = \frac{36\pi V^2}{S^3} \equiv Q_{\rm iso} = \Psi^3, \tag{2}$$

where Ψ is the Wadell sphericity. Equation (2) is the exact analogue of Eq. (1), and it identifies $Q_{\rm iso}$ — not the face count, and not the bond-orientational invariants $W_{12,0}$ or Q_6 — as the correct three-dimensional counterpart of the shape index q.

Table 1: Named structures. η is the packing fraction at contact, $Q_{\text{iso}} = 36\pi V^2/S^3$, z the number of contacts, and "faces" the number of non-degenerate Voronoi faces. Values from explicit half-space construction of the cells.

structure	η	Q_{iso}	faces	z	tang.
FCC	0.740480	0.740480	12	12	yes
HCP	0.740480	0.740480	12	12	yes
FCC − 1-in-13	0.683520	0.683520	11	11	yes
FCC − 1-in-7	0.634698	0.634698	10	10	yes
simple hex., $c = a$	0.604600	0.604600	8	8	yes
FCC − 1-in-5	0.592384	0.592384	9	9	yes
FCC − 1-in-4	0.555360	0.555360	8	8	yes
simple cubic	0.523599	0.523599	6	6	yes
BCC	0.680175	0.753367	14	8	**no**
diamond	0.340087	0.461735	16	4	**no**

The tangentiality hypothesis has a direct physical reading:

every Voronoi neighbour must be a contact.

If a single Voronoi face is supported by a non-touching neighbour, that face lies farther than $\sigma/2$ from the centre, the pyramid decomposition fails, and $\eta \neq Q_{\text{iso}}$. This criterion is the organising principle of what follows, and it immediately partitions the candidate structures of the original work (Table 1).

3 Complete enumeration of tangential Bravais lattices

By Voronoi's theorem the faces of a lattice Voronoi cell correspond to the seven non-zero classes of $L/2L$; a class contributes a non-degenerate face precisely when its minimum norm is attained by the pair $\pm v$ alone. Writing the Selling parameters $p_{ij} = -b_i \cdot b_j$ of a reduced basis b_1, b_2, b_3 with $b_4 = -(b_1 + b_2 + b_3)$, the seven class norms are

$$|b_i|^2 = \textstyle\sum_{j \neq i} p_{ij}, \tag{3}$$

$$|b_1 + b_2|^2 = p_{13} + p_{14} + p_{23} + p_{24}, \tag{4}$$

and cyclic permutations. Each is a *linear* functional of p. Tangentiality — all live class norms equal — is therefore a linear system, and fixing $\sum p_{ij} = 1$ makes the shape space of all three-dimensional lattices the compact 5-simplex $\{p \geq 0\}$. Solving the system exactly over every live-face set and every zero-pattern of p is thus an exhaustive enumeration.

The result is that *exactly three tangential Bravais lattices exist in three dimensions*: the simple cubic lattice (cubic cell, $\eta = \pi/6$), the simple hexagonal lattice at $c = a$ (hexagonal-prism cell, $\eta = \pi/3\sqrt{3}$), and FCC (rhombic dodecahedral cell, $\eta = \pi/\sqrt{18}$).

Two remarks. First, requiring all seven class norms to be equal forces $p_{ij} = 1/6$ by symmetry, whose class norms are 1/2 and 2/3; no fourteen-faced tangential cell exists, so the failure of BCC in Table 1 is structural rather than accidental. Second, the enumeration must be posed as a linear solve. A numerical minimisation of the spread of face distances over the shape space fails, because the objective is non-smooth and because the simple cubic and simple hexagonal solutions lie on boundary strata where several p_{ij} vanish.

4 The depleted-lattice ladder: derivation

De Graaf's two-dimensional states arise by removing particles from the close-packed lattice in a regular pattern. We first exhibit the general structure, then specialise.

Counting identity. Let a contact lattice have coordination z_0, and remove a vacancy set that is *independent* (no two vacancies adjacent) and *uniform*: every remaining particle has exactly K vacant neighbours. Double-counting occupied–vacant edges gives $z_0 N_v = K N_o$, hence

$$\eta_K = \frac{z_0}{z_0 + K}\, \eta_{\rm cp}. \tag{5}$$

For FCC ($z_0 = 12$, $\eta_{\rm cp} = \pi/\sqrt{18}$) this is $\eta_K = 2\sqrt{2}\,\pi/(12+K)$, identical to the ladder found by our sublattice scan with vacancy fraction $1/n$, $n = 12/K + 1$. The identity requires neither congruence nor periodicity; we return to this below.

Independence is forced. Condition (i) is not an assumption. Around an isolated vacancy the surrounding particles are mutually in contact, so the vacated volume is partitioned entirely by contact bisectors and every cell remains tangential. Around a *divacancy* this fails: particles facing each other across the channel acquire Voronoi faces supported at non-contact distance. Numerically, a divacancy in FCC produces a face-distance spread of 0.414 in the adjacent cells, against exactly zero for an isolated vacancy. Tangentiality therefore implies independence.

Spectral bound. A uniform-K independent vacancy set is an equitable 2-partition (perfect 2-colouring) of the contact graph with quotient matrix $\left(\begin{smallmatrix} 0 & z_0 \\ K & z_0-K \end{smallmatrix}\right)$, whose eigenvalues are z_0 and $-K$. On an infinite lattice $-K$ must lie in the Bloch spectrum of the contact graph. The FCC band is $[-4, 12]$, so

$$K \le 4, \tag{6}$$

and the ladder $K \in \{1, 2, 3, 4\}$,

$$\eta_K = \frac{2\sqrt{2}\,\pi}{12+K} = 0.683520,\ 0.634698,\ 0.592384,\ 0.555360, \tag{7}$$

with $z_K = 12 - K$ contacts, is *complete* within this class. Explicit periodic realisations exist for every K: deleting FCC sites with x, y, z all even ($K = 4$), $y + 2z \equiv 0\ (5)$ ($K = 3$), $x + 2y + 3z \equiv 0\ (7)$ ($K = 2$), and $x + 3y + 4z \equiv 0\ (13)$ ($K = 1$, a perfect one-error-correcting code on FCC). All four were verified to be uniform, independent, tangential and congruent, with face vectors 3^8, $3^2 4^7$, 4^{10}, 4^{11}. The double-counting identity and the $K \le 4$ bound were first suggested in an automated review of an earlier version of this work; we verify both and add the spectral formulation, the independence lemma and the results below.

Two dimensions, corrected. The triangular contact graph has band $[-3, 6]$, so $K \le 3$ in two dimensions, and the family is exactly

$$\phi_K = \frac{6}{6+K}\,\frac{\pi}{\sqrt{12}}: \quad \begin{array}{lll} K=3 & \text{honeycomb} & 0.604600 \\ K=2 & \text{kagome} & 0.680175 \\ K=1 & \text{maple-leaf} & 0.777343. \end{array} \tag{8}$$

All three were verified tangential by direct construction, with $\phi = q$ exact. De Graaf's floret pentagonal tiling is the $K = 1$ maple-leaf state and his honeycomb caging state is $K = 3$; the remaining rung, the kagome lattice at $\phi = \sqrt{3}\pi/8 = 0.680175$, is a novel falsifiable prediction lying inside the density range any 2D study already sweeps. An earlier version of this work additionally listed a putative $\phi = 0.453450$ member; that state would require $K = 6 > 3$ and $z = 0$ and is hereby retracted.

Isostaticity. With $z_K = z_0 - K$, the 2D family contains both isostatic numbers exactly: the honeycomb has $z = 3 = d + 1$ (frictional) and the kagome has $z = 4 = 2d$ (frictionless), while the maple-leaf glass state is hyperstatic at $z = 5$. The 3D ladder, $z_K = 8, 9, 10, 11$, contains *neither* 3D isostatic number; $z = 2d = 6$ is instead supplied by the simple cubic lattice of Sec. 3. If rigidity rather than tiling selects the arrest density, two and three dimensions must behave differently.

Degeneracy of the reference states. Because Eq. (5) requires no periodicity, we enumerated *all* uniform-K independent sets on small tori. On a 6×6 triangular torus with $K = 2$ there are four translation-orbits of solutions, of which only one is a lattice coset (the kagome pattern). Checking the non-crystalline states cell by cell: every cell is exactly tangential, every cell has $Q = 0.680175$ to machine precision, and the global density equals $\sqrt{3}\pi/8$ exactly. The geometric reference state at ϕ_K is therefore not a single crystal but a degenerate family containing disordered members. This bears directly on the thermodynamic objection discussed in Sec. 7: whether the degeneracy is extensive is a sharp combinatorial question left open here.

Beyond FCC. The HCP contact graph has Bloch bands spanning $[-4, 12]$, so $-K$ lies in band for all $K \leq 4$: uniform-K states on HCP pass the necessary spectral condition. If they exist they sit at the same densities as Eq. (7) with different cell topologies, so density alone cannot identify the reference structure; the face-adjacency topology must be measured. Finally, the ladder does not exhaust the admissible list: the simple cubic (0.523599) and simple hexagonal (0.604600) lattices of Sec. 3 are tangential but not FCC depletions.

5 Corrections to the three-dimensional discussion

All four items below are reproducible from the accompanying code.

BCC vacancy neighbour count. The original reports "4 nearest neighbours for which the hexagonal face is extended," $\eta = 4\sqrt{3}\pi/35 \approx 0.621874$. A BCC vacancy has *eight* nearest neighbours. Their cells have $V/V_0 = 71/64$, giving $\eta = 8\sqrt{3}\pi/71 = 0.613115$. The six next-nearest values, $V/V_0 = 49/48$ and $\eta = 6\sqrt{3}\pi/49 = 0.666294$, are correct. With eight and six the vacancy volume balances exactly, $8 \times 0.109375 + 6 \times 0.020833 = 1.000000\, V_0$; with the stated counts it does not. Both cell types are non-tangential ($Q_{\mathrm{iso}} = 0.713$ and 0.724 against $\eta = 0.613$ and 0.666), so averaging them to obtain ≈ 0.644 carries no geometric content.

Tetrahedral-octahedral honeycomb. The original states that "each of the two tetrahedra to an octahedron contributes 1/4 of its volume." An octahedron in that honeycomb has eight faces and hence eight adjacent tetrahedra, each shared among four octahedra, contributing two tetrahedra of volume in total rather than one half. The stated cell fills only three quarters of space. Corrected, $\eta = 2\sqrt{3}\pi/27 \approx 0.40307$ rather than $8\pi/(27\sqrt{3}) \approx 0.537422$, and the cell reduces to the FCC rhombic dodecahedron. The reported agreement with frictional random loose packing at $\eta \approx 0.536$ does not survive.

Two-dimensional honeycomb depletion fraction. Section VIII B states that "two out of three particles are removed, with the remaining ones forming a regular honeycomb lattice." Removing two in three leaves a sparser triangular lattice. The honeycomb is one in three removed, consistent with the value $\phi = \pi/(3\sqrt{3}) = (2/3)\pi/\sqrt{12}$ quoted in the same paragraph.

Candidates in the glass range. The original reports that a preliminary search revealed no candidate structure falling in the reported three-dimensional glass-transition range $\eta \approx 0.58$–0.64. Three do: $\eta_3 = 0.592384$, 0.604600 (simple hexagonal at $c = a$) and $\eta_2 = 0.634698$. The first and last coincide, to within the quoted precision, with the avoided mode-coupling density $\varphi_{\mathrm{MCT}} \approx 0.592$ and the relaxation-divergence estimate $\varphi_0 \approx 0.635$ of Berthier and Witten [Phys. Rev. E **80**, 021502 (2009)].

We stress that this last point is not evidence in favour of the hypothesis. The admissible values are dense between 0.52 and 0.74, with typical spacing 0.04, while the literature places the three-dimensional glass transition anywhere in 0.588–0.639. Matching a measured arrest density to a tabulated value is therefore nearly free, and far weaker than the corresponding two-dimensional coincidence, where 0.777 was isolated. Any test must be mechanistic.

6 Numerical test

6.1 Method

We use NVT hard-sphere Monte Carlo with local displacement moves, which provides a stochastic dynamics standing as a proxy for the Brownian dynamics of the colloidal experiments the two-dimensional work targets. Swap moves are used during equilibration and switched off for production. Configurations are initialised on an FCC lattice at the target density with equal radii, melted, and the target size distribution is then imposed and the resulting overlaps annealed to exactly zero. Every final overlap audit returned zero overlapping particles.

$N = 864$; two compositions, a 10% Gaussian polydisperse system and a binary 1:1 mixture at $R = 0.714$; η from 0.40 to 0.62. Structure is analysed with radical Voronoi cells built by half-space intersection.

Two gates were imposed. The equation of state from the contact value of $g(r)$ agrees with Carnahan–Starling to 0.04% at $\eta = 0.30$ (1.4% at 0.40, 4.0% at 0.45; the drift is the linear extrapolation window, not the engine). The Voronoi pipeline reproduces the tabulated Q_{iso} of every perfect tangential lattice to 10^{-8} and returns a contact fraction of 8/14 for BCC.

6.2 Arrest density is estimator-limited

The diffusion coefficient falls four orders of magnitude and the mean-squared displacement exponent collapses from 1.0 to ≈ 0 between $\eta = 0.56$ and 0.60. Fitting $D = A(\eta_a - \eta)^b$ over $\eta \geq 0.48$ gives $\eta_a = 0.6122 \pm 0.0117$ (polydisperse) and 0.6257 ± 0.0121 (binary). These uncertainties are not meaningful: varying the fit window gives 0.599/0.612/0.622/0.884 for the same polydisperse data. The window systematic exceeds the statistical error by an order of magnitude, and the pilot constrains η_a only to roughly 0.60–0.63, which straddles two admissible values and discriminates nothing.

6.3 The contact-fraction signature is degenerate

The most direct probe of the mechanism is the fraction of Voronoi faces that are contacts, which the criterion of Sec. 2 requires to approach unity. As written it is unmeasurable: exact contact has

Table 2: Shell-percolation threshold $\varepsilon^*(\eta)$, in units of the mean diameter, at which the inflated contact network first wraps the box in all three directions. Three configurations per point.

η	polydisperse	binary	rel. diff.
0.40	0.02731	0.02811	2.9%
0.44	0.02154	0.01938	10.6%
0.48	0.01613	0.01565	3.0%
0.50	0.01248	0.01408	12.0%
0.52	0.01179	0.01177	0.2%
0.54	0.00999	0.01006	0.7%
0.56	0.00828	0.00843	1.8%
0.58	0.00679	0.00680	0.1%
0.60	0.00534	0.00496	7.4%
0.62	0.00372	0.00256	36.9%

measure zero in a thermal hard-sphere fluid, and at a strict tolerance the contact fraction is $\sim 10^{-3}$ at every density. Reformulated as $f_c(\delta)$ using the face gap $g_{ij} = (r_{ij} - \sigma_{ij})/\sigma_{ij}$, it is measurable: the median gap falls from 0.248 at $\eta = 0.44$ to 0.054 at $\eta = 0.62$.

However, the gap distribution proves to be single-scale. Rescaling δ by the median gap collapses $f_c(\delta, \eta)$ across all densities: the small-δ ratio $f_c/(\delta/\text{median})$ is $0.79, 0.81, 0.93, 0.84, 0.86, 0.92$ for $\eta = 0.44$ to 0.62, constant within scatter. A single-scale distribution carries no information beyond its own median, and $\mathrm{d}\ln(\text{median})/\mathrm{d}\eta$ steepens monotonically $(-6.4, -7.3, -9.0, -10.5, -13.6)$ with no inflection, cusp or saturation near arrest. The observable is thus not merely null here; it is degenerate with a quantity that varies smoothly through the transition. Any repair must probe the *arrangement* of near-contact faces rather than their number.

Consistently, the mean isoperimetric quotient $\langle Q_{\text{iso}} \rangle$ rises with η and passes through a maximum, at $\eta = 0.60$ (polydisperse) and $\eta = 0.58$ (binary). This is the analogue of the two-dimensional $\max_\phi(n_{\bar{q}})$, which is explicitly the composition-*dependent* feature there; ours likewise moves with composition and is not evidence for a selected state.

6.4 Shell percolation is composition-independent

Table 2 gives the shell-percolation threshold, obtained by bisection with union-find and relative-displacement tracking to detect wrapping clusters. For $\eta \leq 0.60$ the two compositions agree to 4.3% on average, despite representing very different size distributions. Composition independence is, in de Graaf's framework, the signature distinguishing a geometric origin from a packing-detail one, and this is the only observable in the pilot that displays it. The $\eta = 0.62$ binary point deviates by 37% and is almost certainly under-equilibrated near jamming.

The local exponent $\mathrm{d}\ln\varepsilon^*/\mathrm{d}\eta$ is $-5.9, -6.6, -11.0, -7.8, -5.6, -8.8, -9.7, -11.0, -15.0, -18.1$. It steepens overall but wiggles at $\eta = 0.48$–0.52 by more than the quoted spread. With three configurations per point that wiggle cannot be separated from noise: no feature can be claimed, and none can be excluded. Fitting $\varepsilon^* = C(\eta_m - \eta)^p$ returns $\eta_m = 0.83$ and 0.76 with window spreads of 0.15 and 0.13, indicating the assumed form is wrong near contact percolation; restricted windows converge to 0.675 and 0.633, bracketing random close packing.

7 Discussion

The geometry of the three-dimensional problem is now settled. The correct shape observable is Q_{iso}; the admissibility criterion is that every Voronoi neighbour be a contact; there are exactly three tangential Bravais lattices; BCC-derived cells are excluded structurally; and the depleted close-packed states form the discrete ladder of Eq. (7), which subsumes both two-dimensional structures of the original work.

The physics is not settled, and our pilot is not able to settle it. Its useful contribution is negative and methodological. The observable that would most directly test the mechanism is degenerate and shows no feature; the estimator originally proposed for the arrest density is dominated by a fit-window systematic; and the one observable that behaves well, $\varepsilon^*(\eta)$, requires no long-time dynamics at all. This last point substantially reduces the cost of a decisive experiment: short equilibrations at many densities with of order 10^2 configurations each, rather than long dynamical runs, suffice to resolve the local exponent of $\varepsilon^*(\eta)$ and to test whether any feature in it is composition-independent.

The decisive question is accordingly narrow: does the composition-independent $\varepsilon^*(\eta)$ curve possess a resolvable feature, and if so does it lie at one of 0.523599, 0.555360, 0.592384, 0.604600, 0.634698, 0.683520? A negative answer is as informative as a positive one, and given the density of the admissible ladder it must be established mechanistically rather than by numerical coincidence.

Finally, the thermodynamic objection raised against the two-dimensional argument is partially answered by the degeneracy result of Sec. 4: the reference state at η_K is not a single realisation but a family that provably contains non-crystalline members, all exactly tangential at exactly η_K. If the family is extensive the reference states carry configurational entropy, and the mechanism no longer requires a fluid to feel a unique crystal. Whether the count grows exponentially with system size is a sharp combinatorial question, cheap to attack, and in our view the most important open theory item.

Code and data availability

All geometry in Secs. 2–5 is reproduced by `enumerate_3d_ground_states.py` from lattice vectors alone. The Monte Carlo engine, analysis pipeline and percolation code are provided.